

Decomposition-free Synchronous Multi-layer Cooperative 3D Printing for Enhanced Strength and Structural Integrity

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Abstract

Cooperative 3D printing offers unique opportunities for the manufacturing of large-scale structures by integrating multiple printing heads to facilitate fast, multi-material, or multi-resolution printing. However, most existing cooperative strategies rely on decomposing the part into sub-domains assigned to individual heads, which alters inter- and intra-layer gap times and introduces weak interfaces at decomposition boundaries. These effects modify the material bonding and the stress distribution within the deposited material, thereby compromising mechanical performance and increasing the risk of failure. To address these challenges, a novel decomposition-free Synchronous Multi-Layer Printing (SMLP) strategy is proposed. In SMLP, each head prints an entire layer rather than a segment, with heads operating under a controlled time shift to ensure synchronous deposition paths. This approach is designed to combine the mechanical advantages of single-head printing with much of the build-time efficiency of multi-head systems. To validate the effectiveness of this approach, investigations on thin wall specimens built via two types of extrusion-based additive manufacturing process is conducted. First, SMLP is implemented on a dual-head Fused Deposition Modeling system to perform tensile and tear tests on extrusion-based polymer printing, comparing the resulting bond strength against traditional multi-head strategy. Second, to assess implications in large-scale applications, an analytical model for extrusion-based concrete printing is developed that couples time-dependent yield stress evolution with shear stress distribution. Experimental results indicate that SMLP delivers superior interlayer tensile strength compared to traditional multi-head printing strategy (~30% higher strength than traditional multi-head) while preserving tear resistance. Analytically, the strategy reduces build time by approximately 24% relative to single-head printing and increases the number of buildable layers before collapse by 100–300% compared to traditional multi-head strategies. Together, these findings provide a unified framework for designing cooperative printing strategies that improve both strength and structural integrity along with productivity.

Keywords: Cooperative 3D printing, Multi-head printing; Stress analysis; Analytical modeling; Part decomposition

1. Introduction

Large-scale additive manufacturing (AM) is gaining popularity among researchers and industry professionals for autonomous fabrication of large components such as full-size prototypes of submarine hulls and multi-story buildings [1,2]. It opens new avenues for building infrastructures in harsh environments or in space with minimal to no human involvement. Existing commercially available large-scale 3D printers [3,4] are capable of working with a diverse range of materials, including composites, cementitious materials, and metals. More recently, cooperative 3D printing involving robot swarms or teams of robots has demonstrated increased mobility and flexibility compared to traditional gantry-based systems, expanding applications within AM [5,6]. The adoption of multiple print heads in cooperative 3D printing enables the manufacturing process to be decomposed into smaller tasks that can be performed in parallel, leading to higher efficiency and faster build speeds [7]. Moreover, it enables the realization of functionally-graded, multi-material, and/or multi-resolution parts for enhanced properties and performance [8,9].

At the same time, multi-head printing introduces new complexities: different print strategies, parameters, and toolpaths change the inter- and intra-layer gap time (the waiting time between successive passes of the nozzle), the extent of bonding or fusion, and the stress distribution within the deposited material. These effects ultimately control the quality, strength, and lifespan of the printed structure. Traditionally, multi-head printing has relied on decomposing a part into sub-sections, with each print head working on a distinct sub-part for cooperative 3D printing [10,11]. However, this structural decomposition often leads to bonding issues at the interfaces between sub-parts, which are typically the primary failure locations, resulting in reduced structural integrity and degraded dimensional accuracy [12]. Various solutions, including angular decomposition strategies and non-flat interfaces based on topological interlocking geometries, have been proposed to improve bonding at sub-part boundaries [13,14]. Despite these advances, current approaches still struggle to achieve seamless, non-decomposed structures (as illustrated in Figure 1); as a result, multi-head printed parts often fall short of the structural quality achieved with single-head printing. These limitations indicate that current multi-head approaches mainly improve productivity at the potential expense of structural performance. There is little guidance on how cooperative strategies should be designed so that they preserve, or even enhance (if possible), the mechanical integrity achieved by conventional single-head printing [15]. More specifically, to date, there is no decomposition-free, mechanically informed cooperative printing strategy—or accompanying modeling framework—that enables multi-head printing to achieve structural performance comparable to or possibly exceeding that of traditional single-head or cooperative printing, particularly for thin-wall, large-scale structures.

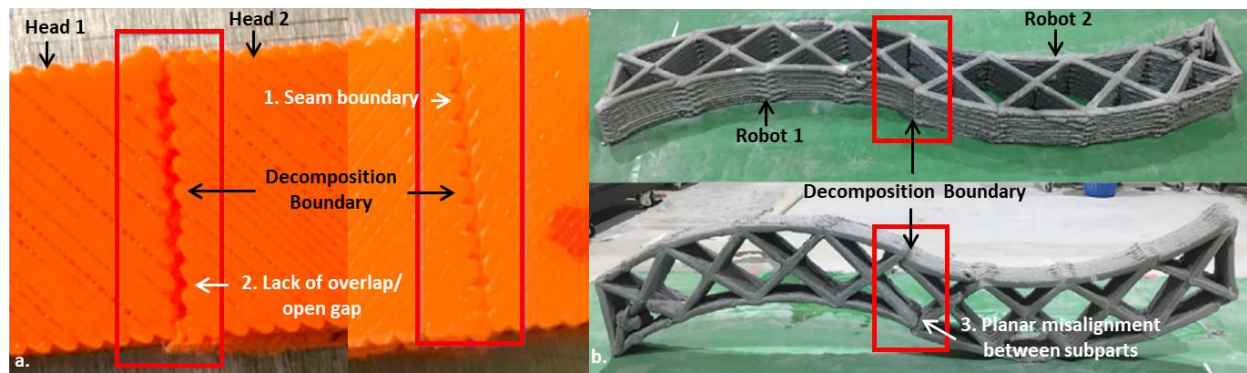


Figure 1. Illustrative examples of the inferior bonding and quality at decomposition boundary in existing multi-head cooperative printing scenarios, potentially leading to points of structural failure: a. Extrusion-based polymer cooperative printing [13], and b. Extrusion-based concrete cooperative printing [16].

To fill the above gap and address this trade-off between productivity and structural performance, this study introduces and evaluates a novel, decomposition-free multi-head cooperative printing strategy named synchronous multi-layer printing (SMLP). Unlike traditional "concurrent" strategies that print different segments of the same layer, SMLP assigns each head to print a complete layer. The heads operate simultaneously on consecutive layers with a controlled time delay, allowing them to follow one another in a synchronized path as shown in Figure 2. Few researchers have defined the traditional decomposition-based multi-head strategies as "concurrent" or "simultaneous" 3D printing, where multiple heads print different segments of the same layer concurrently [16]. However, this is different from our proposed strategy, in which multiple heads are simultaneously printing consecutive layers. This approach is designed to combine the build-time efficiency of multi-head systems with the mechanical integrity of single-head approaches by maintaining a continuous deposition path without vertical seams.

In this work, our overall objective is to understand how the proposed SMLP cooperative multi-head strategy would influence both the mechanical performance and deformation behaviour of extrusion-based printed structures (specially large-scale structures with a focus on thin-wall geometries), compared to traditional cooperative multi-head and single-head strategies. For example, large-scale construction features are generally produced by extrusion-based concrete AM process, which requires a careful examination of potential quality risks under multi-head cooperative print strategies. Existing studies have primarily examined single-head systems [17], where relationships between rheology [18–20], buildability, toolpath design [21–23] and failure modes are now relatively well understood [24,25]. However, for multi-head systems, the toolpath and inter-pass timing are fundamentally altered, which changes the evolving stress state in the deposited material and can induce new failure modes that are not captured by single-head models [26]. At the same time, direct experimental exploration of these strategies, especially at architectural scales, is costly, time consuming, and often impractical [27]. This creates a clear need for modeling tools that can rapidly evaluate multi-head strategies and provide process guidelines without extensive trial-and-error.

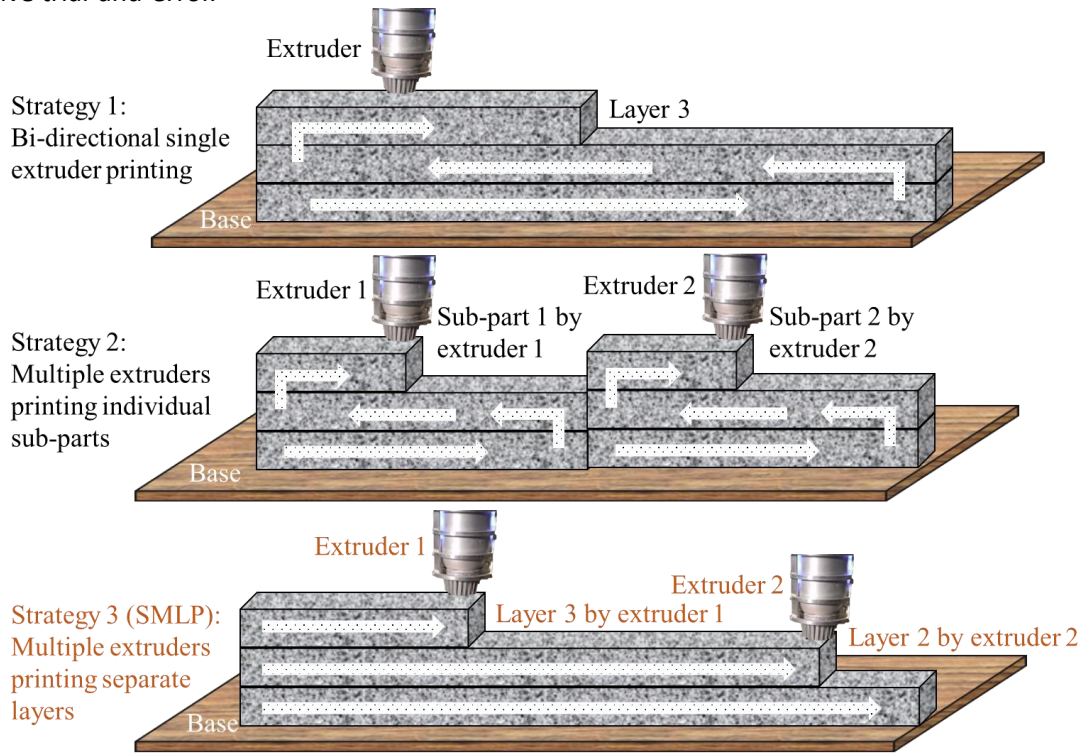


Figure 2. Simple illustration of different single-head and multi-head strategies adopted in this work, with the newly proposed Strategy 3 to enhance structural integrity.

Addressing the above objective, we pursue two complementary studies in extrusion AM that are linked by a common physical idea: failure occurs when the stresses generated during printing exceed the evolving strength of the material, whether tensile strength in a polymer or yield stress in fresh concrete. First, we use a polymer material on a dual-extruder fused deposition modeling (FDM) system to experimentally quantify interlayer bond strength by performing tensile and tear tests on thin wall specimens. This setup enables direct measurement of how cooperative multi-head strategies, together with print speed and layer length, influence mechanical properties. Experimental studies that compare cooperative multi-head strategies with polymer material remain very limited, so this work provides rare quantitative data on the process-structure-property relationship. Second, we develop an analytical framework for extrusion based concrete printing in which a time dependent yield stress law and the stress build up during printing are

used to predict deformation and collapse for thin walls. The maximum shear stress distribution during printing using Murata's slump prediction model [19] and force equilibrium, and the local yield stress is estimated from the elapsed time after deposition. The goal is to demonstrate SMLP (Strategy 3 in Figure 2) as a superior multi-head strategy compared to the traditional decomposition-based multi-head strategy (depicted as Strategy 2 in Figure 2), while highlighting print speedups compared to single-head Strategy 1 in Figure 2. In the polymer experiments, we quantify how SMLP modifies inter and intra layer gap times and measure the resulting tensile and tear strengths for different speeds and lengths. For concrete AM, we compare the predicted stress and yield stress fields for these different strategies to estimate the time of collapse of deposited concrete. Total build time and collapse time are used as key performance metrics, and a sensitivity analysis on rheological properties and printing parameters provides additional insight into structural stability and failure.

In summary, this work introduces a novel decomposition-free cooperative SMLP strategy, provides systematic experimental evidence from extrusion-based polymer AM on its effect on mechanical properties of thin-wall specimens, and develops an analytical framework for extrusion-based concrete AM that links multi-head toolpaths to early-age collapse, together offering a unified basis for designing structurally robust cooperative printing strategies.

2. Methodology

The printing path is one of the critical input parameters that typically must be determined before printing, including specifying the start and end coordinates as well as the trajectory and motion planning (e.g., for robotic systems). Even for a simple thin-wall geometry in a single-head additive manufacturing scenario, multiple path-planning strategies may be adopted, such as bi-directional versus single-directional deposition. Different printing paths result in different total build times and inter-layer gap times, which leads to different stress distribution within the deposited material. This consideration becomes even more important with multiple print heads, as a larger number of interconnected printing strategies become available. Traditionally, the part is decomposed into n sub-parts, which are then assigned to n printing heads [2,10]. For example, as illustrated Figure 2, two print heads collaborate to print a wall by dividing it into two sections, with each section managed by a separate print head. Alternatively, the SMLP approach proposed in this study assigns each layer of the part to a different head for printing, enabling coordinated printing of consecutive layers without geometric decomposition.

In this section, we present the methodology for two complementary studies in extrusion-based additive manufacturing of thin-wall specimens: (i) extrusion-based printing of polymeric materials and (ii) extrusion-based printing of concrete materials. These studies are designed to systematically evaluate how different cooperative printing strategies, particularly the proposed SMLP approach, affect build time, inter-layer timing, stress evolution, and resulting mechanical performance.

2.1 Experimental procedure for extrusion-based polymer printing

2.1.1 Materials and equipment

The filament used in this study is Bambu PLA Basic, a modified polymer composite consisting of 99% polylactic acid and less than 1% of additives such as Polycaprolactone (PCL), this is a common blend formula, oriented for high-speed printing and can achieve printing speeds up to 250-300 mm/s in certain machines. Being the most common material in FDM printing, PLA is widely adopted as it's easy to print and inexpensive. All samples were printed on a modified Independent Dual Extruder (IDEX) machine based on Tronxy Gemini S. As shown in Figure 3c, a customized dual-head mount was utilized for SMLP capabilities and it had two separate control boards for each hotends compared to the in-built capability.

2.1.2 Specimen and experimental design

For the experimental testing conducted in this study, ASTM D882-18 (similar to ISO 527-3 but not considered fully equivalent) was adopted. This standard specifies the determination of tensile properties for plastics in the form of thin sheets less than 1.0 mm in thickness, a condition compatible with the capabilities of the experimental setup. The nominal width of the specimens shall be in the range from 5.0 mm to 25.4 mm, and the width to thickness ratio should be greater than 8. And the test specimens should be selected so that thickness is uniform to within 5% of the average thickness over the length of the specimens between the grips. Specimens had a height of 18 mm (60 deposited layers at 0.3 mm) and a thickness of 0.4 mm. One of the objective was to evaluate the effect of SMLP printing strategy for cooperative printing on the tensile strength and inter-layer adhesion. The independent variables to achieve this where varied as follows: print speed: 5 mm/s to 60 mm/s with 5 mm/s step size and length: 50 mm, 100 mm and 150 mm. The parameters that remained unchanged were layer height: 0.3 mm, gap between two heads: 33 mm, travel speed: 60 mm/s, nozzle temperature: 208°C and bed temperature: 60°C. For each combination of print speed and length, a total of two replicate specimens were fabricated. Additionally, to demonstrate the effectiveness of proposed strategy, it was compared with the traditional decomposition-based multi-head cooperative printing strategy, as illustrated in Figure 2b. Prior to mechanical testing, all specimens underwent visual inspection. Samples with noticeable defects, such as flowrate fluctuations, insufficient inter-layer adhesion, or visible void disconnected regions, were identified and excluded from further analysis. Valid specimen was then utilized for tensile strength and tear strength analysis, assessing resistance to force applied along the specimen's width.

2.1.3 Testing procedure

Tensile testing was performed using a TSA750 test stand with a M5-750 digital force gauge and G1061-2 wedge grips, the combination provides capacity upto 3750 N (750 lbf / 375 kgF), Accuracy of the force gauge is $\pm 0.1\%$ of full scale, and resolution is 1/5000. Specimens were mounted in the grips with the longitudinal axis aligned with the direction of applied force. Constant incremental force was applied gradually until specimen rupture. The crosshead speed was selected to provide a consistent strain rate according to the guidelines. Prior to testing, specimen dimensions were verified using vernier calipers with an accuracy of ± 0.02 mm. Specimens exhibiting obvious manufacturing defects or failures at the grip interface (jaw breaks) were excluded from the dataset. The second replicate from each experimental condition was subjected to tear resistance analysis. The specimen was oriented to apply load perpendicular to the deposited layers to assess the resistance to force applied along the specimen's width. The maximum force required to initiate and propagate separation was recorded as the tear strength.

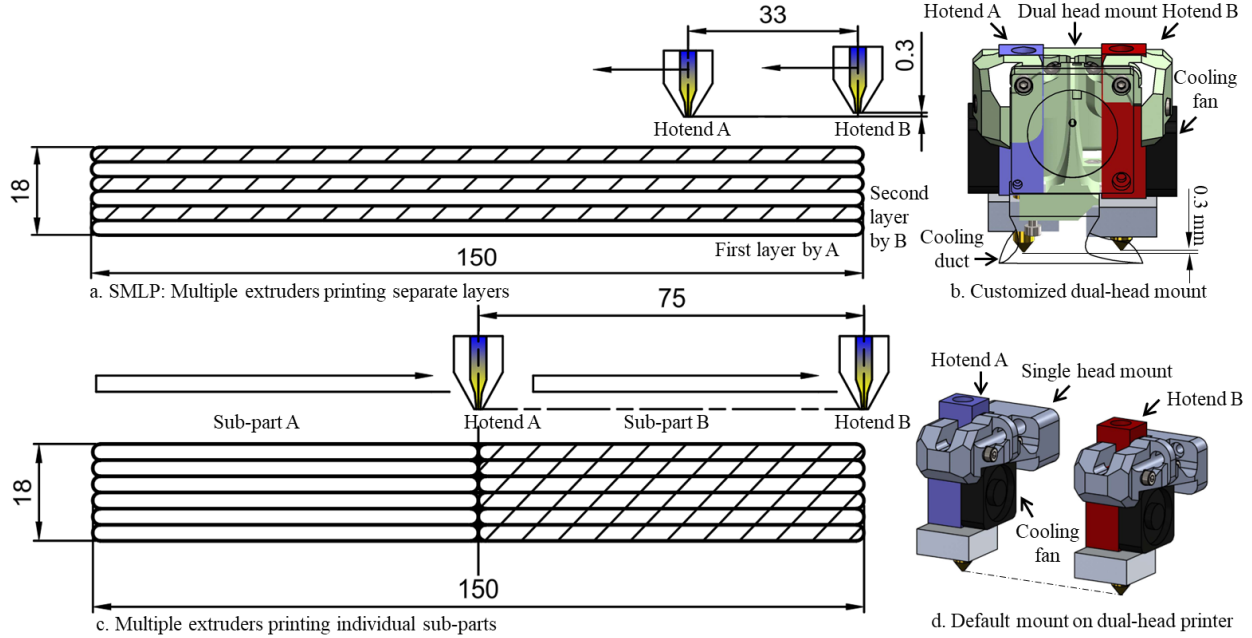


Figure 3. Illustration of two printing strategies for multi-head printing utilized in the experiment a. SMLP and c. Bi-directional multi-head strategy, with b. customized dual-head mount for SMLP printing and d. default mount for dual-head printing on the FDM machine.

2.2 Analytical modeling for extrusion-based concrete printing

2.2.1 Assumptions for the analytical model

The structural integrity of printed structures fabricated using the single-head and multi-head strategies (traditional versus the proposed approach) are evaluated through analytical stress analysis associated with extrusion-based additive manufacturing of concrete material for a straight wall. In this process, it is widely known that the elapsed time from the deposition of fresh concrete to a fully printed structure affects the stress distribution within the part, and thus, adequate yield stress of fresh concrete to print is necessary to avoid any deformation or collapse during the printing process. Consequently, analytical models are established for the newly presented cooperative printing strategy to detect early failure, and are then compared with the existing results associated with single-head and traditional multi-head cooperative printing strategies. Figure 4 shows the flow chart illustrating the analytical framework utilized in this study. With design requirements and printing strategies considered as inputs, the analytical model iteratively evaluates each layer and discrete points throughout the layer, starting from layer 1, to determine if the yield stress of the printed concrete is higher than the maximum shear stress, ensuring a stable printing process. If maximum shear stress exceeds the yield stress in the initial layers, the model indicates early deformation and predicts the time of collapse, which is inspired from a verified model [27]. Important metrics such as deformation and build time for all strategies are obtained from the analytical model. Based on these metrics, an optimal strategy can be decided for cooperative 3D printing. Several assumptions were made to simplify the calculations:

1. The interlayer bond strength reduces as the printing time gap between layers increases. However, Chen et al. [30] demonstrated that, in large-scale printing with single head, short time intervals (<10 minutes) had minimal to no effect on the bonding properties based on experimental and numerical studies. Moreover, for parts of the same size, multi-head strategies typically entail shorter printing gap times between layers compared to single-head printing. As a result, since the

time gap between layers in this study is less than 10 minutes, the impact of printing gap times on the bonding can be neglected, and thus, only stress-driven phenomena and their effect on structural integrity are considered.

2. The supply of raw materials is assumed to be continuous and ideal.
3. Fresh concrete is deposited vertically, with the width, height (i.e., thickness), and length of each layer remaining constant throughout the study. A multi-layer, single-track structure is adopted.
4. The extrusion head is assumed to deposit perfectly rectangular cross-section beads.
5. The speed of multiple extrusion heads are assumed to be similar and constant throughout this study.

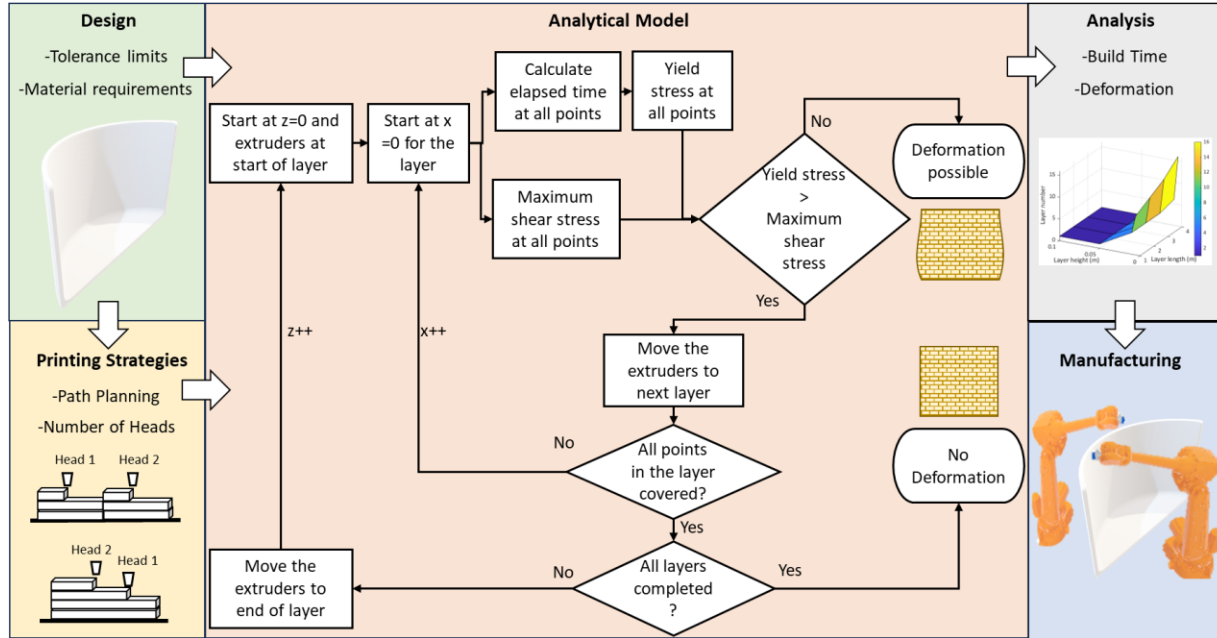


Figure 4. Analytical framework for modeling structural integrity in extrusion based concrete AM of large-scale thin-wall structures.

2.2.2 Yield stress distribution

Studies in the literature have demonstrated that the yield stress of fresh concrete varies with elapsed time (t_p) in either linear or exponential manner [25]. This relationship can be expressed as follows:

$$\tau_x(t_p) = \tau_0 e^{\alpha t_p} \quad (1)$$

$$\tau_x(t_p) = \alpha t_p + \tau_0 \quad (2)$$

where α and τ_0 are the hardening coefficient and the initial yield stress. Equation 1 or 2 can be selected and applied based on the yield stress measurement results of the fresh concrete. However, to ensure safety, it is always assumed that the yield stress calculated from the elapsed time-yield stress equation is lower than the measured yield stress [27].

2.2.2.1 Single-head printing (Strategy 1)

In Strategy 1, at any point where fresh concrete has been extruded at time (t_0), and with the current position of the extruder (t_e), the yield stress can be calculated as follows:

$$\tau_{1x}(L_e, L_o) = \tau_0 e^{\alpha \left(\left(\frac{h}{V_t} \right) \left(\left[\frac{L_e}{l} \right]_f - \left[\frac{L_o}{l} \right]_f \right) + \frac{L_e - L_o}{V_p}} \right) \quad (3)$$

$$\tau_{1x}(L_e, L_o) = \alpha \left(\left(\frac{h}{V_t} \right) \left(\left[\frac{L_e}{l} \right]_f - \left[\frac{L_o}{l} \right]_f \right) + \frac{L_e - L_o}{V_p} \right) + \tau_0 \quad (4)$$

where L_e and L_o are the respectively lengths measured from extruder position at any point (o) from start point, as illustrated in Figure 5a. In the above equations, the yield stress is expressed as a function of lengths. However, it is preferable to represent yield stress as a function of the spatial coordinates (x, z) . This is because the coordinates provide a clearer understanding of the locations of potential failures. Therefore, the following relationship can be established:

$$L_o(x, z) = \begin{cases} x + l \left[\frac{z}{h} \right]_f & , \text{where } \left[\frac{z}{h} \right]_f \text{ is odd} \\ (l - x) + l \left(\left[\frac{z}{h} \right]_f - 1 \right) & , \text{where } \left[\frac{z}{h} \right]_f \text{ is even} \end{cases} \quad (5)$$

$$L_e(x', z') = \begin{cases} x' + l \left[\frac{z'}{h} \right]_f & , \text{where } \left[\frac{z'}{h} \right]_f \text{ is odd} \\ (l - x') + l \left(\left[\frac{z'}{h} \right]_f - 1 \right) & , \text{where } \left[\frac{z'}{h} \right]_f \text{ is even} \end{cases} \quad (6)$$

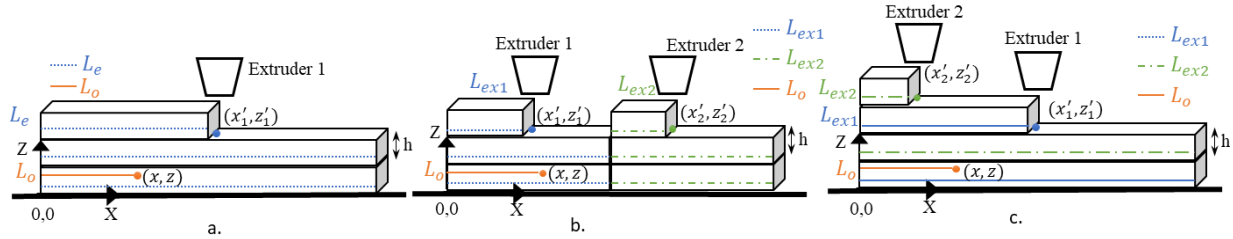


Figure 5. Different lengths based on current extruder position for a. Strategy 1 (single-head), b. Strategy 2 (traditional decomposition-based multi-head), and c. Strategy 3 (proposed SMLP).

2.2.2.2 Cooperative multi-head printing using traditional multi-head strategy (Strategy 2)

For Strategy 2, at any point where fresh concrete has been extruded at time (t_0) , and with the current position of the extruders to be similar to that in strategy 1, and given that the elapsed time for the material extruded by both the extruders is consistent, the shear stress can be calculated as follows:

$$\tau_{2x}(L_o, L_e) = \tau_0 e^{\alpha \left(\left(\frac{h}{V_t} \right) \left(\left[\frac{2L_e}{l} \right]_f - \left[\frac{2L_o}{l} \right]_f \right) + \frac{L_e - L_o}{V_p} \right)} \quad (7)$$

$$\tau_{2x}(L_o, L_e) = \alpha \left(\left(\frac{h}{V_t} \right) \left(\left[\frac{2L_e}{l} \right]_f - \left[\frac{2L_o}{l} \right]_f \right) + \frac{L_e - L_o}{V_p} \right) + \tau_0 \quad (8)$$

However, as a function of coordinates, the location of the extruders plays a crucial role in establishing the relationship between shear stress distribution across the entire built wall. The extruders operate in independent regions without overlaps. To determine which extruder was used to deposit material at any point (o), one can simply check if its x coordinate is between 0 and $l/2$ or between $l/2$ and l as shown in Figure 6.

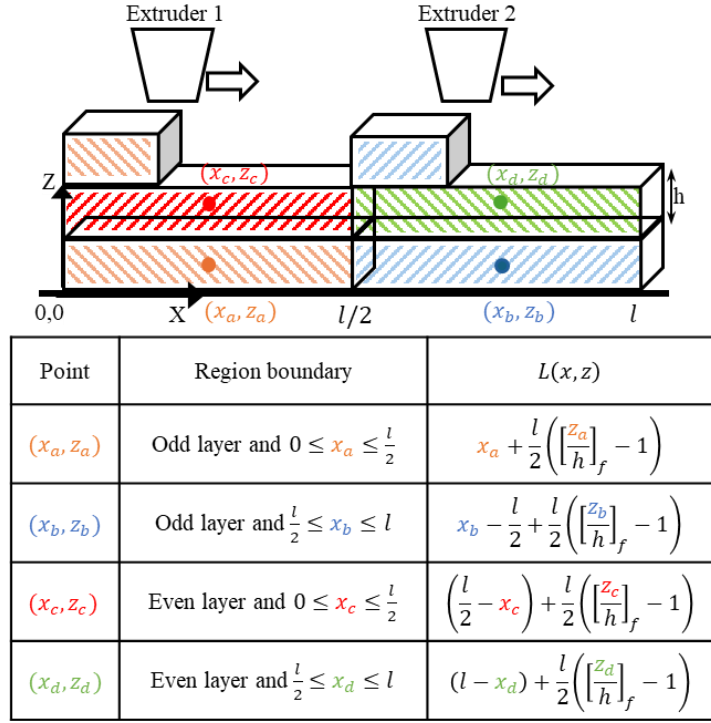


Figure 6. Relationship between various points in the printing path to the layer length in strategy 2.

2.2.2.3 Cooperative multi-head printing using SMLP (Strategy 3)

For SMLP, at any point where fresh concrete has been extruded at time (t_0), and with the current position of the extruder, the yield stress can be calculated, as shown:

$$\tau_{3x}(L_o, L_{ex1}, L_{ex2}) = \begin{cases} \tau_0 e^{\alpha^* \left(\left(\frac{D}{V_{max}} \right) \left(\left\lceil \frac{L_{ex1}}{l} \right\rceil - \left\lceil \frac{L_o}{l} \right\rceil \right) + \frac{L_{ex1} - L_o}{V_p} \right)} & , \text{where } \left\lceil \frac{L_o}{l} \right\rceil_c \text{ is odd} \\ \tau_0 e^{\alpha^* \left(\left(\frac{D}{V_{max}} \right) \left(\left\lceil \frac{L_{ex2}}{l} \right\rceil - \left\lceil \frac{L_o}{l} \right\rceil \right) + \frac{L_{ex2} - L_o}{V_p} \right)} & , \text{where } \left\lceil \frac{L_o}{l} \right\rceil_c \text{ is even} \end{cases} \quad (9)$$

where L_{ex1} and L_{ex2} are the lengths measured for extruder positions and L_o is the length measured for any point (o) from start point of the extruder used for deposition as shown in Figure 5c. The time delay does not affect the elapsed time because, depending where the layer point (o) exists, it is compared with the respective extruder's position. For instance, if point (o) is in an odd layer, the elapsed time is calculated with extruder one's position. The yield stress as a function of the coordinates of point (x, z) in the space can be established as:

$$L_o(x, z) = \begin{cases} x + l \left(\frac{\left\lceil \frac{z}{h} \right\rceil - 1}{2} \right) & , \text{where } \left\lceil \frac{z}{h} \right\rceil_c \text{ is odd} \\ x + l \left(\frac{\left\lceil \frac{z}{h} \right\rceil - 1}{2} \right) & , \text{where } \left\lceil \frac{z}{h} \right\rceil_c \text{ is even} \end{cases} \quad (10)$$

$$L_{ex1}(x'_1, z'_1) = x'_1 + l \left(\frac{\left\lceil \frac{z'_1}{h} \right\rceil - 1}{2} \right); \quad L_{ex2}(x'_2, z'_2) = x'_2 + l \left(\frac{\left\lceil \frac{z'_2}{h} \right\rceil - 1}{2} \right) \quad (11)$$

Where $\lceil \cdot \rceil_c$ is the ceiling function and (x'_1, z'_1) and (x'_2, z'_2) are the extruder positions for extruder 1 and 2, respectively.

2.2.3 Maximum shear stress distribution

The relationship between material properties or rheological properties of concrete and stress is well established. The pressure exerted by the weight of concrete above a layer results in a compressive normal stress on that layer. This compressive normal stress due to height z of material printed on the n^{th} layer in an x - z plane can be shown as [27]:

$$\sigma(z) = \rho g(nh - z) \quad (12)$$

where ρ is the density of fresh concrete and g is the gravity acceleration. The maximum shear stress which acts on the surface is approximated by one half of the pressure and can be expressed as:

$$\tau_{\max(z)} = 0.5\rho g(nh - z) \quad (13)$$

The maximum shear stress distribution for a fully formed part with the same number of layers will be consistent across different strategies. However, when fresh concrete is only partially formed within the same layer, the maximum shear stress distribution will vary depending on the strategy employed.

2.2.3.1 Single-head printing (Strategy 1)

For strategy 1, the normal compressive stress and the maximum shear stress generated at a position (x', z') within a build-up layer in the x - y plane can be calculated as shown in Figure 7, where τ_d is the shear stress caused by the discharge pressure. It is an additional stress acting at the point where the printing is taking place. There are three primary regions, i.e., before, at and after the head, that have the change in normal compressive and maximum shear stress of the deposited concrete.

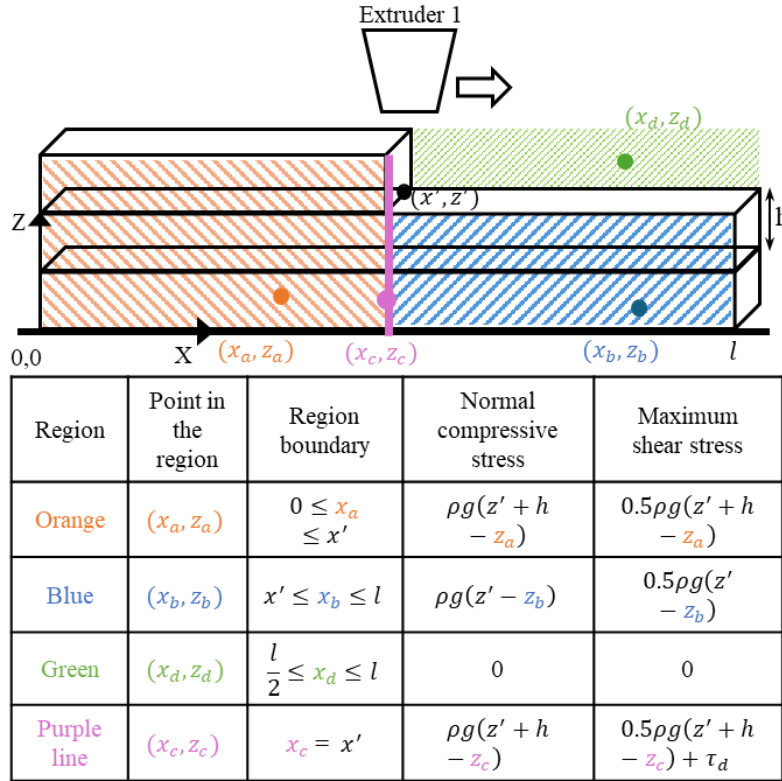


Figure 7. Normal compressive and maximum shear stress while printing using strategy 1.

2.2.3.2 Cooperative multi-head printing using traditional multi-head strategy (Strategy 2)

For strategy 2, with two extruders located at (x'_1, z'_1) and (x'_2, z'_2) , where z'_1 and z'_2 are equal as both are printing the same layer. Additionally, extruder head 2 will be $l/2$ units away in the x-direction from extruder head 1. The normal compressive stress generated from these positions of a build-up layer in the x-y plane is shown in Figure 8 with consideration of various scenarios. The scenarios are critical as the printing direction is changed across layers, which impacts the deformation or failure point of the structure. Moreover, the maximum shear stress generated from these positions of a build-up layer in the x-y plane as illustrated in Figure 8.

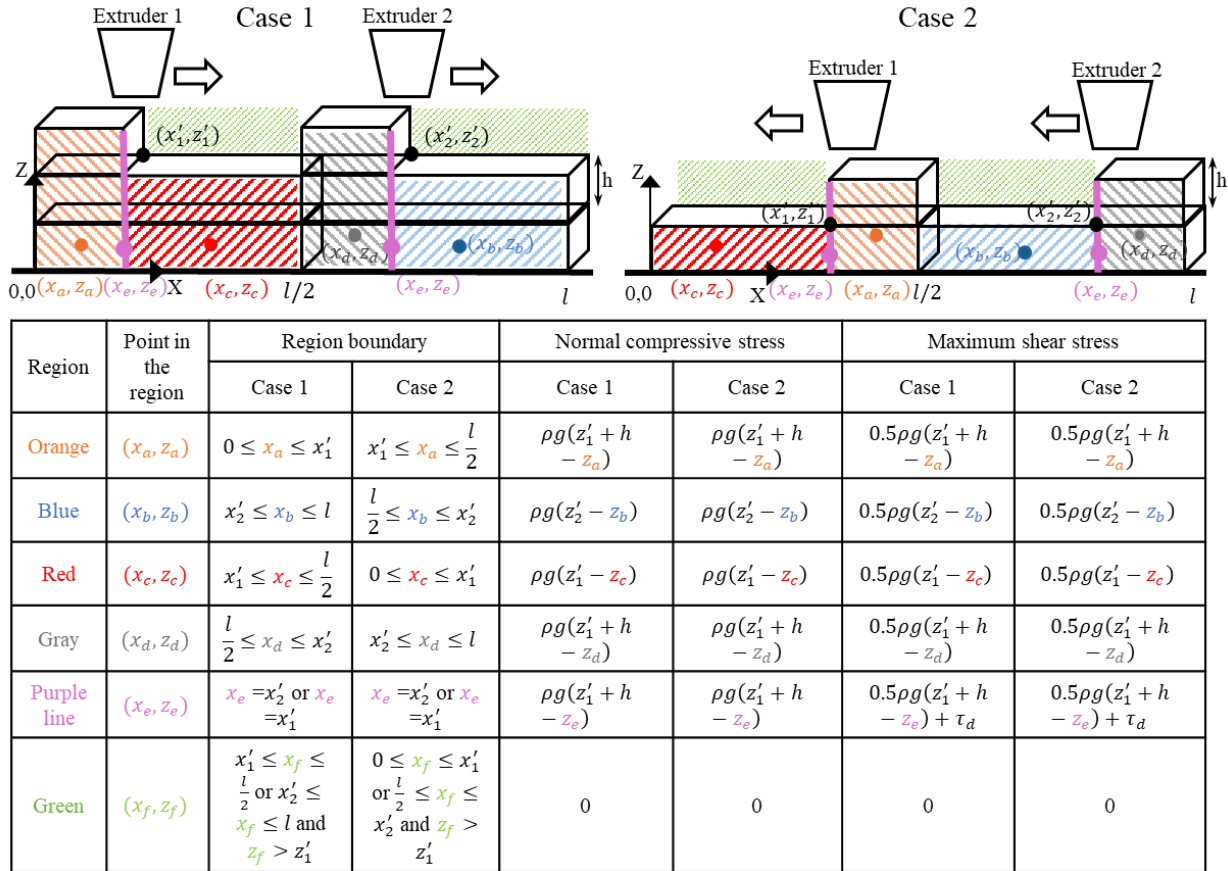


Figure 8. Normal compressive and maximum shear stress while printing using strategy 2

2.2.3.3 Cooperative multi-head printing using SMLP (Strategy 3)

In Figure 9, τ_{d1} and τ_{d2} represent the shear stresses caused by the discharge pressures of extruder 1 and extruder 2. It is important to note that z'_1 is consistently used in the tables as both z' s, (i.e., z'_1, z'_2) are the same for uniform layer heights. However, in SMLP, there are two extruders operating at different z-positions (x'_1, z'_1) and (x'_2, z'_2) , as illustrated in Figure 5c. Therefore, from a positional perspective, two configurations are possible: either extruder 1 is ahead of extruder 2, or vice versa. The relative position of the extruders impacts the failure point of the wall. Additionally, this relative position is influenced by several factors, including speed of extruders, the time delay between the operations of extruders, and the number of layers to be printed. The normal compressive stress and maximum shear stress generated at specific positions within a build-up layer in the x-y plane for SMLP can be calculated as illustrated in Figure 9, where τ_{d1} and τ_{d2} are the shear stress caused by discharge pressures of extruder 1 and extruder 2.

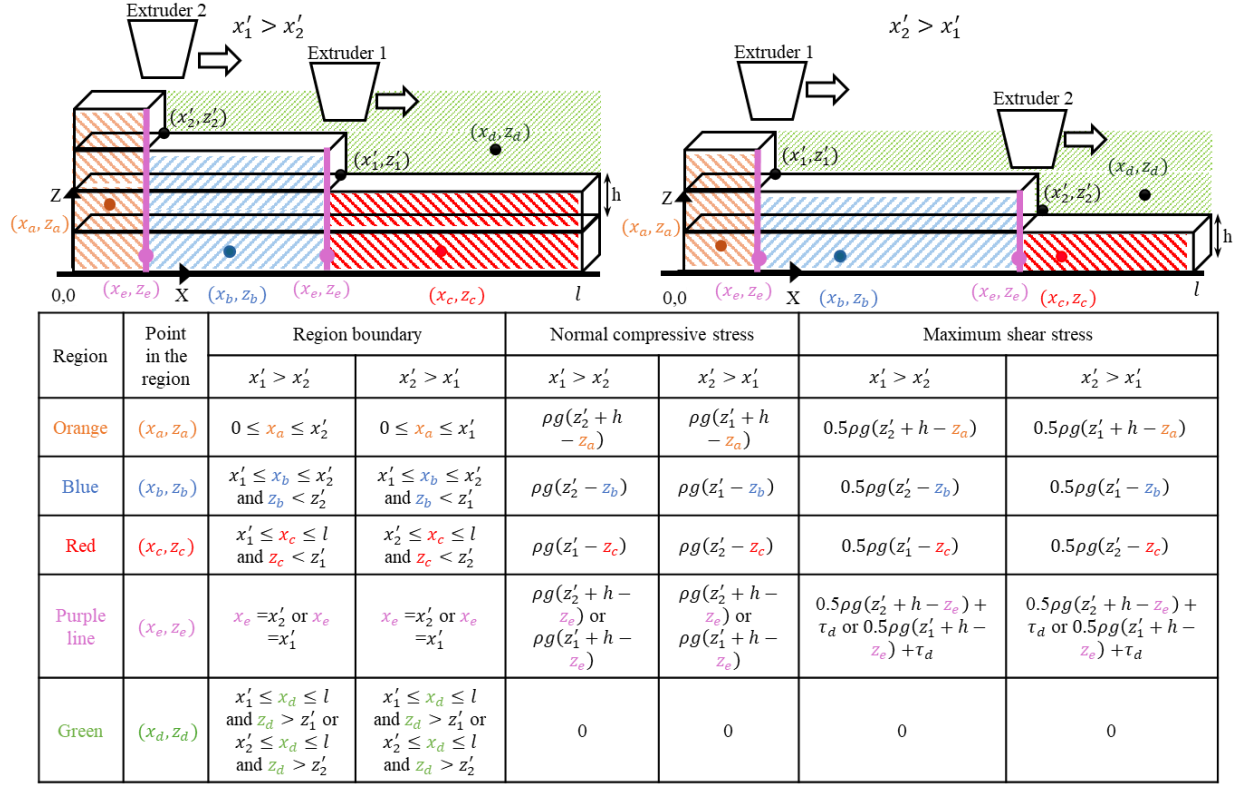


Figure 9. Normal compressive and maximum shear stress while printing using SMLP strategy.

3. Results

3.1 Experimental results for extrusion-based polymer printing

3.1.1 Analysis for synchronous multi-layer printing

The material was initially tested along the vertical height of thin wall. A two-way ANOVA indicates both factors significantly affect tensile strength: print speed, $F(11,22)=2.52$, $p=0.031$; layer length, $F(2,22)=4.58$, $p=0.022$ as shown in Table 1. Model fit is moderate ($R^2=62.6\%$) with residual standard deviation $S=1.29$. From sums of squares, the effect size is larger for speed than for length: partial $\eta^2_{\text{speed}} = 46.17/(46.17+36.61) \approx 0.56$ vs partial $\eta^2_{\text{length}} = 15.23/(15.23+36.61) \approx 0.29$. The probability test on the data showed normality (Anderson–Darling test, $AD = 0.384$, $p = 0.377$) and all the assumptions of ANOVA were met (shown in Figure 19 in the appendix) Practically, this says (i) where you set the speed matters more for tensile performance than which of the tested length you print, and (ii) once speed is controlled, length still contributes a non-trivial ($\approx 30\%$) share of explainable variance.

Table 1. Two-way ANOVA results for tensile strength.

Source	DF	SS	MS	F	P
Print speed	11	46.16	4.20	2.52	0.031
Layer Length	2	15.23	7.62	4.58	0.022
Error	22	36.61	1.66		
Total	35	98.01			

Later, the thin wall was tested along it's horizontal length. For tear strength, both main effects are again significant and stronger than for tensile: print speed and layer length as displayed in Table 2. Model fit is higher ($R^2=74.7\%$) with residual $S=1.88$. Effect sizes from sums of squares show speed dominates: partial

$\eta^2_{\text{speed}} = 188.355/(188.355+77.476) \approx 0.71$ (large), while partial $\eta^2_{\text{length}} = 40.270/(40.270+77.476) \approx 0.34$ (moderate). From an engineering perspective, the interpretation is that tear resistance is even more speed-sensitive than tensile strength, highlighting the dominant role of inter-pass gap time in interfacial bonding. At very low speeds, excessive aging of the previously deposited material weakens the interface, whereas at very high speeds insufficient wetting and compaction limit bonding. Layer length remains a meaningful secondary factor, primarily through its influence on inter-pass delay. The tear strength data was approximately normally distributed as per the normality test (Anderson–Darling test, $AD = 0.377$, $p = 0.392$). Figure 20. Residual plots for validating assumptions for ANOVA testing for tear strength. Figure 20 in the appendix shows that all the assumptions for ANOVA were satisfied.

Table 2. Two-way ANOVA results for tear strength.

Source	DF	SS	MS	F	P
Print speed	11	188.35	17.12	4.86	0.0001
Layer Length	2	40.270	20.14	5.72	0.0010
Error	22	77.476	3.52		
Total	35	306.10			

3.1.2 Comparative results from experiments

Paired t-tests were used to compare the mechanical performance of thin-wall specimens printed using the traditional decomposition-based multi-head strategy and the proposed SMLP strategy. For tensile strength, SMLP exhibits a statistically significant improvement over the traditional multi-head approach (mean gap $\approx +12.4$ in SMLP's favor, 95% CI [10.7, 14.1], $p < 0.001$), while no statistically significant difference is observed in tear strength between the two strategies (mean gap -0.27 , 95% CI $[-2.88, 2.34]$, $p = 0.831$), as shown in Table 3 and Table 4, respectively. For paired t-test with null hypothesis being that the difference between traditional multi-head and SMLP is 10 was statistically significant which is supported by the mean gap of 12.4 MPa (30% higher than traditional multi-head). Additionally, Figure 10 supports the t-test results that SMLP has higher tensile strength with similar tear strength as traditional multi-head printing. In other words, SMLP significantly improves tensile strength without compromising tear strength. Together with the ANOVAs, the picture is consistent: (i) choose speed first, because it is the largest lever for both tensile and tear; (ii) use SMLP within the mid-speed window, because it preserves tear strength and substantially elevates tensile strength while enabling faster builds; and (iii) account for layer length as a secondary lever via its impact on inter-pass gap time.

Table 3. Paired t-test for tensile strength of thin wall via traditional multi-head and SMLP.

Source	N	Mean	Stdeviation	SE mean
Bi-directional	12	35.07	2.55	0.74
SMLP	12	47.49	1.16	0.33

Table 4. Paired t-test for tear strength of thin wall via traditional multi-head and SMLP.

Source	N	Mean	Stdeviation	SE mean
Bi-directional	12	13.88	2.98	0.86
SMLP	12	14.15	3.17	0.91

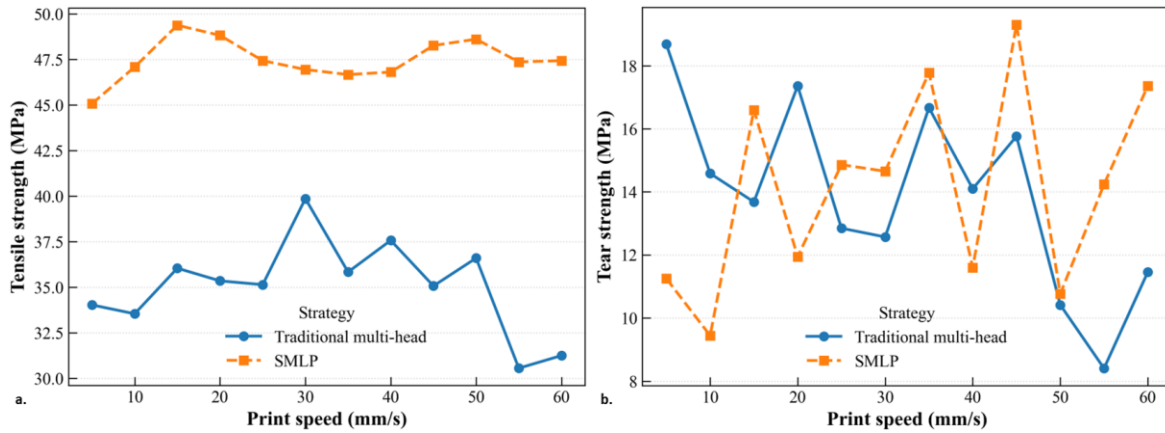


Figure 10. Comparison of a. tensile strength and b. tear strength of parts printed via bi-directional (traditional multi-head) and SMLP printing strategies

Another point of discussion is initiation and the propagation of cracks for both the printing strategies. As the scope of this study is to introduce the novel printing strategy and investigate its strength compared to traditional strategy, we briefly discuss about the failure of specimens and the initiation of the crack. As clearly seen in Figure 11, for the traditional multi-head printing strategy, the tensile test failure occurs at the decomposition boundary, which proves that the weakest spot is placed on the decomposition boundary.

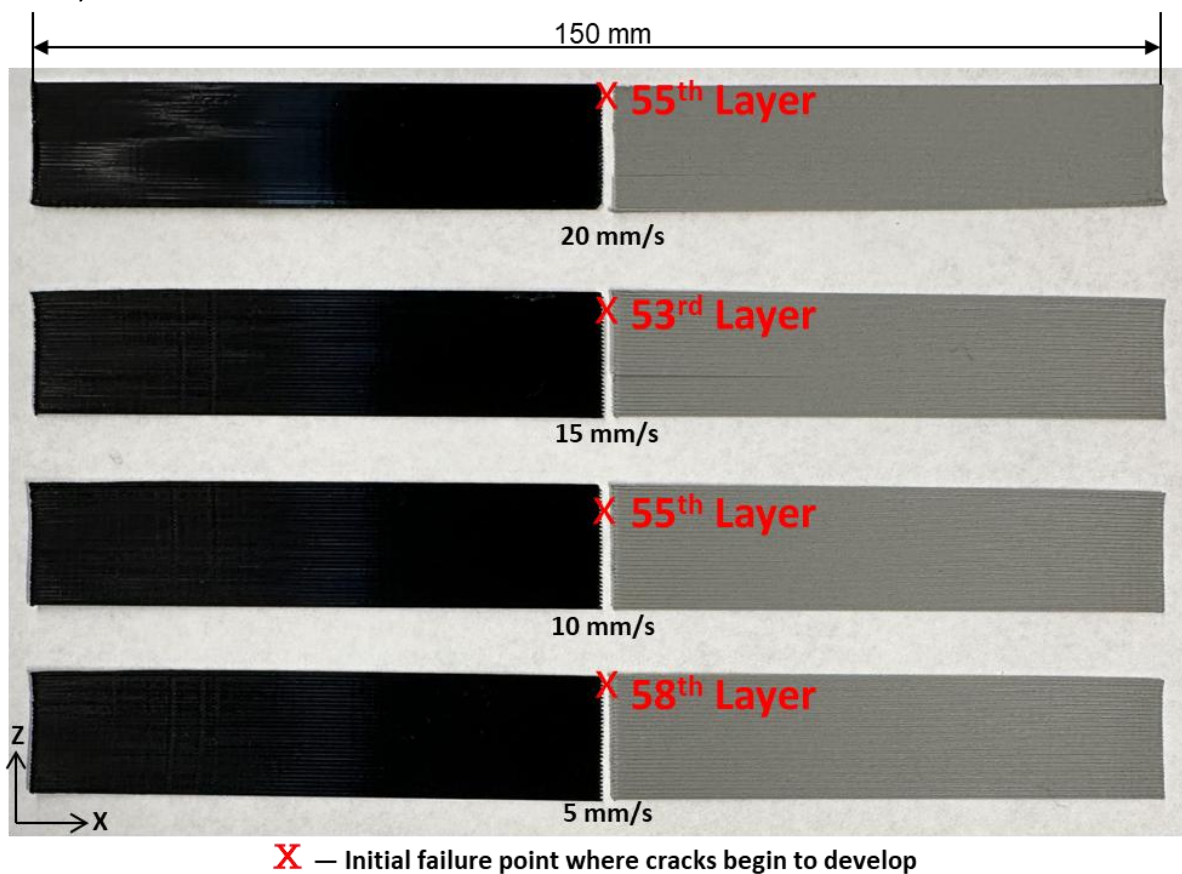


Figure 11. Failure location observed during tensile testing of specimens printed using the traditional multi-head strategy, showing that fracture consistently initiates at the decomposition seam.

3.2 Simulation results for extrusion-based concrete printing

3.2.1 Analysis for synchronous multi-layer printing

To test the hypothesis that SMLP strategy enhances the printing efficiency compared to single-head strategy (Strategy 1) while simultaneously enhancing the structural integrity over traditional multi-head strategy (Strategy 2), two complementary metrics are evaluated. build time is compared between SMLP and the single-head strategy to quantify efficiency gains. Second, the number of layers deposited before collapse is compared between SMLP and the traditional multi-head strategy to assess improvements in structural stability. As demonstrated in Figure 12, SMLP strategy enables the construction of substantially more layers prior to collapse than Strategy 2 (as many as 60 additional layers), while also achieving a significant reduction in the total build time in comparison to Strategy 1. Overall, SMLP increases the number of printable layers by approximately 100–300% compared to the traditional multi-head strategy and reduces build time by approximately 24% compared to the single-head strategy.

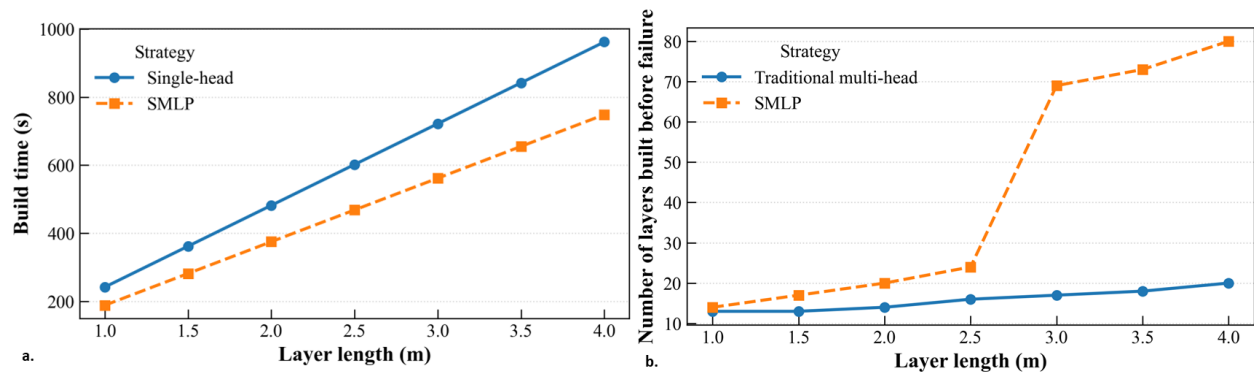


Figure 12. Performance of SMLP compared to a. strategy 1 in build times and b. strategy 2 in number of layers built.

The influence of various rheological properties and printing parameters on the structural integrity of concrete deposited through the SMLP strategy was analyzed using analysis of variance (ANOVA). The factors examined were initial yield stress (A: low - 500 Pa, high - 4000 Pa), time gap (B: low - 0.5 s, high - 5.0 s), layer length (C: low - 0.5 m, high - 4.0 m), layer height (D: low - 0.01 m, high - 0.1 m), and print speed (E: low - 25 mm/s, high - 100 mm/s), with each factor tested at two levels. The response variable was defined as the time of failure, measured by the number of layers.

ANOVA was conducted up to second-order terms, as shown in Table 5. Based on the p-values obtained, initial yield stress and layer height emerged as the most significant linear terms, while the interaction terms AD and CD were the most significant quadratic terms. The time gap between the two heads had a relatively minor effect on the structure's failure time; however, minimizing this time gap could improve efficiency. Therefore, it is recommended to minimize the distance/time gap between the heads, provided it does not lead to collisions. In subsequent sections, the impact of these significant terms on both structural integrity and build time metrics will be compared with those of two other strategies.

Table 5. ANOVA analysis results for SMLP for time of structural failure.

Source	DF	SS	MS	F	P	
A - Initial yield stress	1	7938	7938	36.51	0	Significant
B - Time gap	1	1.1	1.1	0.01	0.944	
C - Layer length	1	288	288	1.32	0.267	
D - Layer height	1	4560.1	4560.1	20.97	0	Significant

E - Print speed	1	392	392	1.8	0.198	
AB	1	0.1	0.1	0	0.981	
AC	1	242	242	1.11	0.307	
AD	1	3321.1	3321.1	15.27	0.001	Significant
AE	1	338	338	1.55	0.23	
BC	1	1.1	1.1	0.01	0.944	
BD	1	0	0	0	1	
BE	1	120.1	120.1	0.55	0.468	
CD	1	1596.1	1596.1	7.34	0.015	Significant
CE	1	200	200	0.92	0.352	
DE	1	351.1	351.1	1.61	0.222	
Error	16	3479	217.4			
Total	31	22828				

3.2.2 Comparative results from analytical model

The length of wall is fixed at 0.15 m for the proposed analytical model, allowing verification with the results from literature model [27]. The parameters selected for this study are detailed in Table 6.

Table 6. Case study parameters and their values [27].

Parameter	Symbol	Value
Initial yield stress	τ_0	500 Pa
Hardening co-efficient	α	0.0003/s
Acceleration of gravity	g	9.8 m/s ²
Density of concrete	ρ	2500 kg/m ³
Traverse speed	V_t	0.01 m/s
Print speed	V_p	0.025 m/s
Layer height	h	0.01 m

Across all three strategies, the maximum shear stress surpasses the yield stress of the printed concrete as the extruder initiates printing layer 5. In other words, during the printing of layer 5, deformation becomes evident at layer 1, as depicted in both Figure 13 and Figure 14. These figures illustrate the evolution of yield stress and maximum shear stress during the construction process using strategies 2 and 3, respectively. It is worth noting that these figures illustrate the yield stress distribution across the entire deposited surface for both Strategy 2 and the SMLP method, with sub-figures capturing this distribution at distinct time intervals. Deformation is apparent in both strategies when layer 6 is completely printed. The yield stress shows a little inclination at the top because material deposited in the upper layers experiences a delayed deposition relative to the lower layers, leading to lower yield stress values in the upper layers compared to the base layers. Consequently, the top layers exhibit progressively lower yield stress at these specific intervals, while the base layers continue to build yield stress over time.

Another important metric for simulation model is the computation time. The analytical model took only 0.0181 seconds for one run. Furthermore, a comparison for total build time underscores the efficacy of these multi-head strategies. For 5 layers with a length of 0.15 m, the total build time is 38, 20, and 30.33 seconds for strategies 1, 2, and 3, respectively. Notably, the wall begins deforming at layer 5 across all

strategies, indicating that the adoption of multiple heads has not affected the start time for collapse. In fact, the utilization of multi-heads has led to a reduction in the overall build time for the wall.

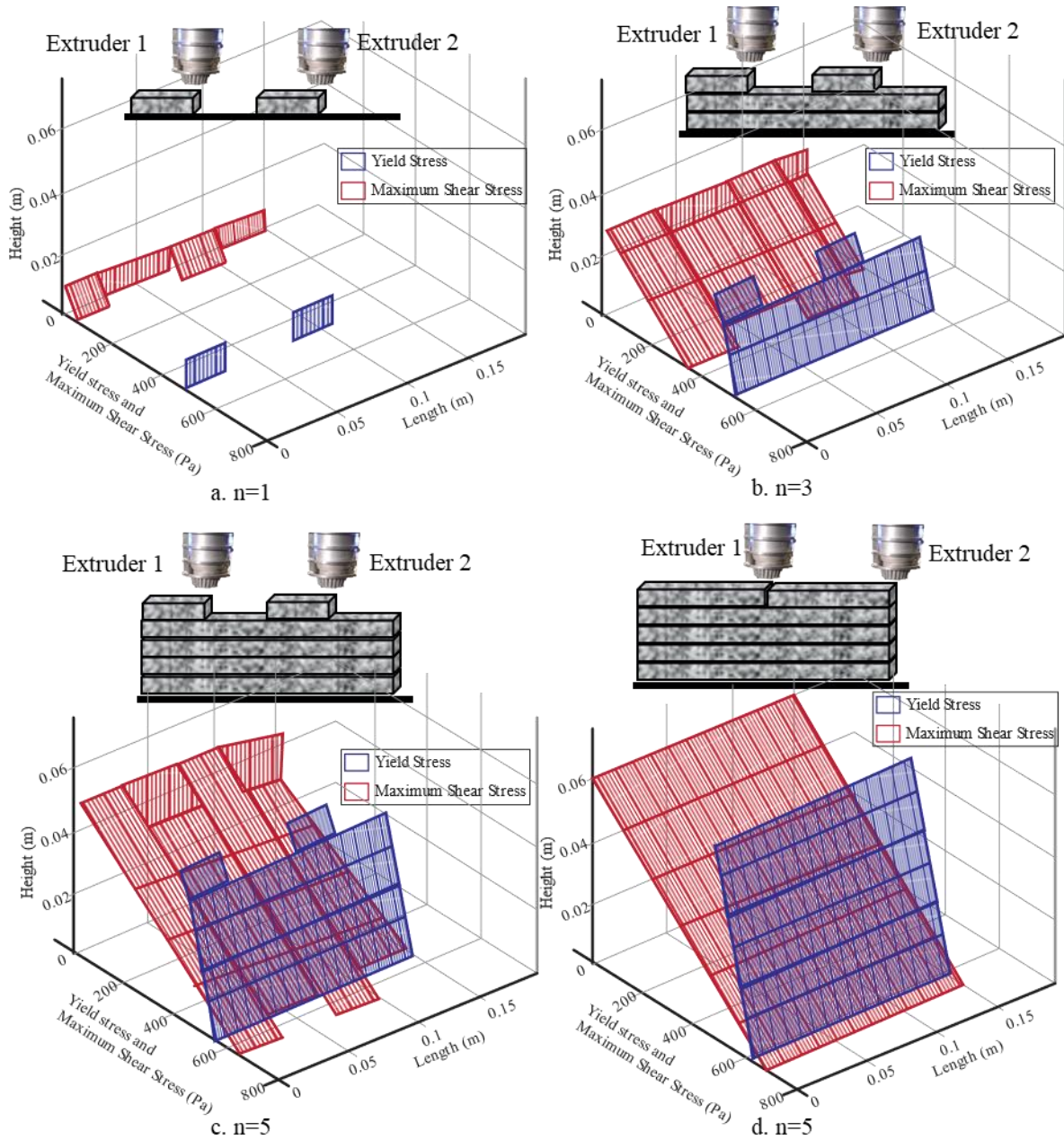


Figure 13. Comparison of yield stress and maximum shear stress for strategy 2 a. $n=1$, b. $n=3$, c. start of $n=5$ and d. completion of $n=5$ layer.

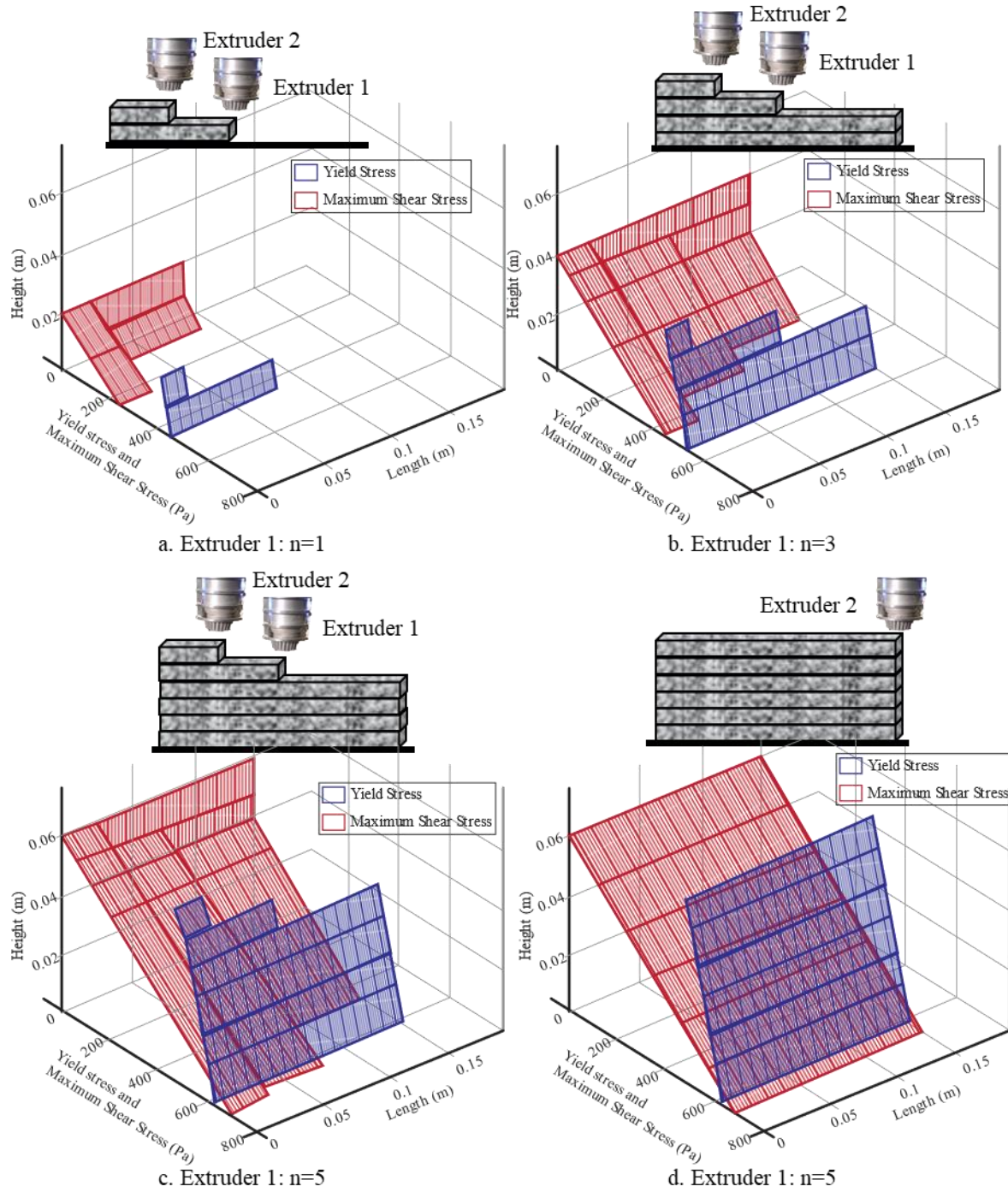


Figure 14. Comparison of yield stress and maximum shear stress for SMLP a. Extruder 1 $n=1$, b. Extruder 1 $n=3$, c. Extruder 1 $n=5$ and d. completion of Extruder 1 as well as extruder 2

3.3.3 Sensitivity Analysis

3.3.3.1 Impact of initial yield stress of concrete

The rheological properties of materials are fundamental in determining the deformation and buildability of parts. Specifically, the initial yield stress of fresh concrete is a controllable parameter that can be enhanced during its preparation. Various methods, such as incorporating accelerators or other additives

into the fresh concrete, can be employed to achieve a higher initial yield stress [18]. According to the findings presented in Section 3.2.1, an increase in the initial yield stress from 500 to 4000 Pa [31,32] affects the time to collapse, as depicted in Figure 15. By elevating the initial yield stress, fresh concrete can bear greater weight and sustain higher build layers before structural failure occurs. Specifically, increasing the initial yield stress from 500 to 4000 Pa resulted in a roughly 600% increase in the number of layers sustained before failure across all strategies. In other words, more layers could be successfully built. Among the strategies, the proposed SMLP strategy (Strategy 3) demonstrated a slight advantage ($\sim 3.03\%$) over Strategy 1 (single-head print), which itself showed better performance ($\sim 6.06\%$) than Strategy 2 (traditional multi-head print strategy). Furthermore, the benefits of higher initial yield stress became increasingly apparent as the yield stress rose, as shown in Figure 15b. This indicates an improved stress distribution in the proposed cooperative printing approach compared to the conventional single-head and multi-head printing strategies.

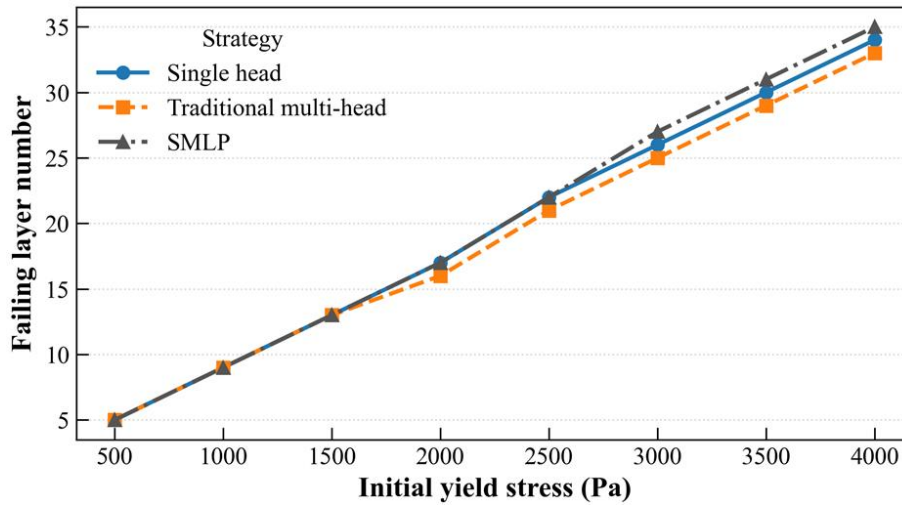


Figure 15. Sensitivity analysis for varying initial yield stress on a. failing layer number (i.e., time of collapse) for all three strategies.

3.3.3.2 Impact of print speed and layer length

The extrusion head speed and layer length were adjusted to assess their influence on both build time and the yield stress of the deposited concrete. Equations to calculate the total build time for all the strategies are provided in the Appendix. In this study, three different speed values, i.e., 25 mm/s, 50 mm/s, and 100 mm/s, were employed. The number of layers was held constant at 5, with a 1-second time gap between the two extruders. In Figure 16, it can be seen that SMLP saves time compared to strategy 1 and is slower in comparison to strategy 2. As illustrated in Figure 17a, the yield stress development for SMLP very closely follows that of strategy 1, which is widely used in the industry. In addition to that, SMLP achieves higher yield stress value compared to strategy 2 as seen in Figure 17b. Furthermore, among the strategies examined, SMLP fell between strategies 2 and 1 in terms of both build time and yield stress development metrics. Notably, SMLP exhibited closer alignment with strategy 1 in terms of yield stress development and approached strategy 2 in the build time metric. This highlights the advantages inherent in the proposed multi-head printing strategy, making it favorable in real-world applications.

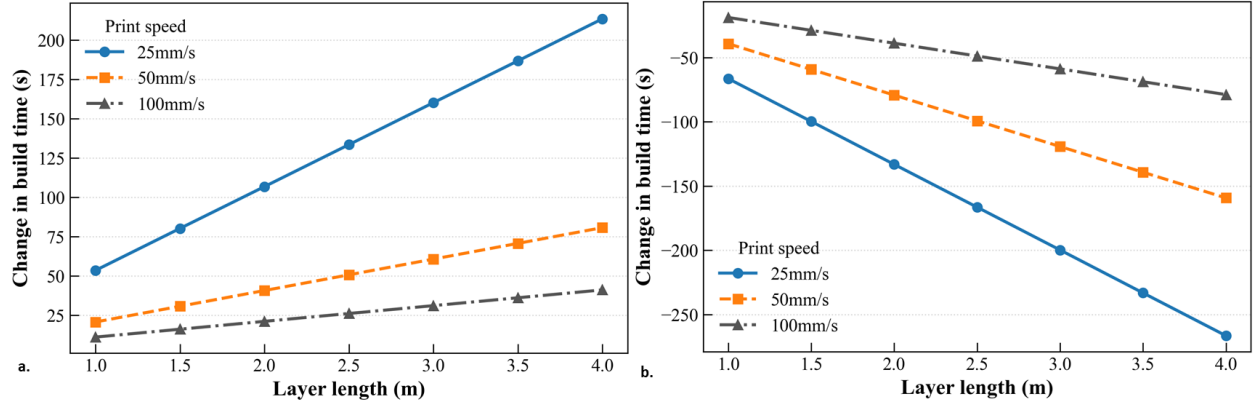


Figure 16. Influence of print speeds on the performance of SMLP compared to a. strategy 1 and b. strategy 2 in build times.

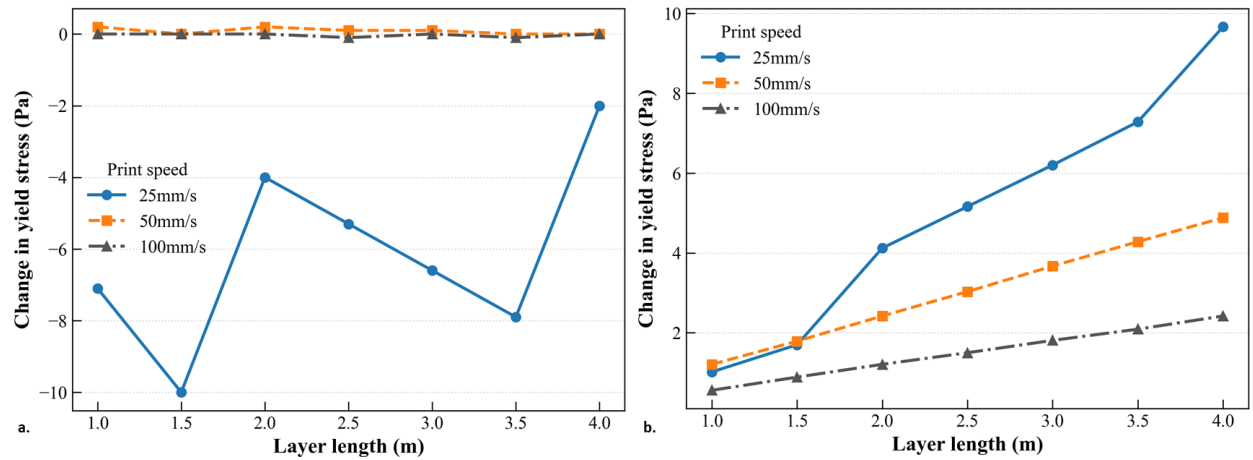


Figure 17. Influence of print speeds on the performance of SMLP compared to a. strategy 1 and b. strategy 2 in yield stress.

3.3.3.3 Impact of layer height

The investigation into collapse times for all three strategies across varying layer heights, with a fixed print velocity of 25 *mm/s*, was explored, as illustrated in Figure 18. As the layer height increases, there is a corresponding increase in the maximum shear stress on the initial layer, resulting in premature failure. Hence, a smaller layer height is preferable across all strategies, regardless of layer length. Among the strategies evaluated, strategy 1, which utilizes a single head, exhibits superior performance across all combinations of layer heights and lengths compared to other strategies. For instance, with a layer length of 4 *m* and a layer height of 0.01 *m*, strategies 1, 2, and 3 can print 20, 12, and 16 layers, respectively, before any indication of collapse. In other words, SMLP can build 25% more layers before failing compared to strategy 2. Furthermore, it is noteworthy that the SMLP approach can print a greater number of layers compared to strategy 2 for walls of equivalent length and height.

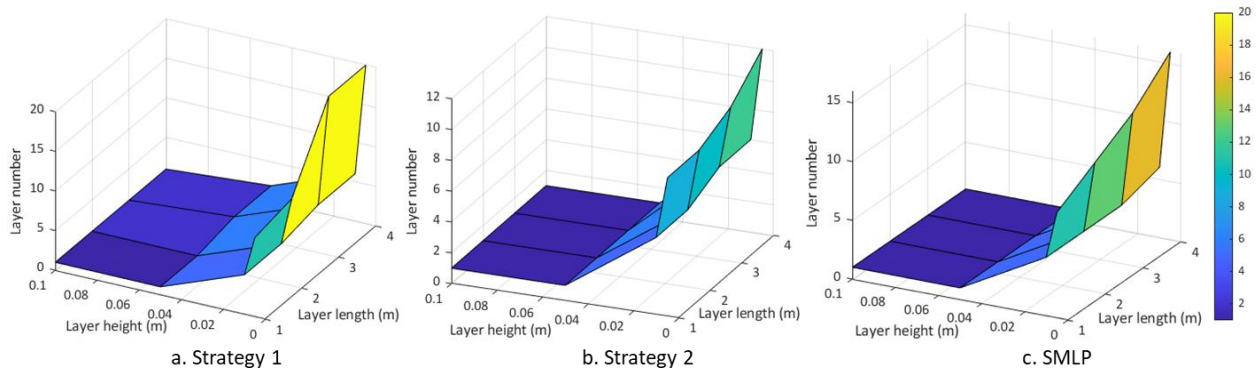


Figure 18. Time of collapse for varying layer height/thickness and layer length for a. strategy 1, b. strategy 2 and c. SMLP.

4. Limitations

The applicability of SMLP is highly dependent on the motion system used. If a traditional gantry-based system is utilized, the printed geometry is generally limited to hollow structures to allow one print head to follow the other without restriction. Complex geometries may require more advanced setups like drones or 6-DOF robotic arms. The implementation on gantry-based system requires specific hardware modifications, such as a customized dual-head mount and separate control boards for each hotend to enable the synchronized delay. SMLP requires higher computational power compared to single-head strategies, specifically to manage collision avoidance between multiple active heads. Coordinated layer-wise assignment requires accurate synchronization of two or more heads in space and time and continuous collision avoidance, especially when the heads operate at different heights and pass over previously printed material. On real hardware this translates into more complex motion planners, tighter requirements on calibration, and potentially higher sensitivity to tracking error or communication delays. However, this challenge exists for most of the multi-head printing strategies and not SMLP. Moreover, as the number of heads increases the spacing in height between heads shrinks, collision risk and coordination complexity grow and non-productive travel may increase if all heads must span the entire layer. At some number of heads, the additional speed gain may not justify the extra complexity, especially on gantry systems where all heads share same axes. In future studies, these limitations can be addressed.

5. Conclusion and Future Work

In this study, a novel decomposition-free cooperative printing strategy, named synchronous multi-layer printing (SMLP), is proposed. In SMLP each printing head is assigned to complete individual layers in a coordinated manner, so that consecutive layers are deposited with controlled and nearly uniform inter and intra layer gap times. This design aims to retain the mechanical robustness typically associated with single head printing while achieving build times comparable to conventional multi-head strategies.

The proposed strategy was first implemented on a dual head FDM system using PLA, and its performance was compared with a conventional multi-head strategy. Tensile and tear tests on printed walls covering a range of printing speeds and layer length showed that SMLP can increase strength and reduce variability. SMLP significantly improves tensile strength by roughly 12 MPa (which is approximately 30% higher than strength of traditional multi-head) without degrading tear strength. These experiments provide rare quantitative evidence that cooperative multi-head strategies, when designed to control the material at bonding, can improve mechanical performance rather than simply matching single head behavior.

To investigate the implications for extrusion based concrete printing, an analytical model was formulated that couples a time dependent yield stress law with the evolving shear stress distribution in walls printed using SMLP, decomposition based multi-head printing, and single head printing. Collapse was predicted as the time when the maximum shear stress exceeds the local yield stress. The analytical model supports the hypothesis that SMLP can substantially delay collapse and reduce deformation relative to traditional strategies. For the parameter ranges considered, SMLP increased the number of layers that could be built before failure by roughly 100 to 300 percent compared with the decomposition based strategy, and reduced build time by about 24 percent relative to bi directional single head printing. The parametric study further showed that initial yield stress and layer height have the largest influence on collapse time for all strategies. SMLP benefits particularly from higher initial yield stress, where its advantage over other strategies becomes more pronounced. Across strategies, higher initial yield stress, moderate printing speeds that avoid excessive inter pass waiting times, smaller layer heights, and appropriately chosen wall lengths were identified as desirable for enhancing structural integrity.

Overall, the combined experimental and analytical results demonstrate that cooperative printing strategies do not have to trade structural performance for productivity. By explicitly accounting for the interaction between toolpath design, gap times, and material build up, SMLP provides a pathway to faster and more reliable large scale additive manufacturing. Future work will extend the framework to more complex geometries and loading scenarios, and further tune coordination among multiple print heads to target specific structural performance and quality requirements. Additionally, future work will investigate different decomposition patterns for effect on the mechanical properties and will study microstructural behavior including crack initiation and propagation in more detail.

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Appendix

For strategy 1, if the velocity of nozzle while printing and traversing is V_p and V_t , respectively. The print time for layer i is L_i/V_p and traverse time between layer i to layer $i + 1$ would be L_t/V_t , where L_i and L_t

are the length of printing path and traverse length, respectively. So, the total build time for strategy 1 can be written as:

$$TBT_1 = \sum_i \left(\frac{L_{p,i}}{V_p} \right) + \frac{(n-1)*h}{V_t} \quad (14)$$

where h is the layer height and is the traversed length by the extruder head in the strategy 1. For strategy 2, the total build time for strategy 2 with m robots can be written as:

$$TBT_2 = \sum_i \left(\frac{L_{p,i}}{mV_p} \right) + \frac{(n-1)*h}{V_t} \quad (15)$$

In this work, as two extruders are used, the equation will become as,

$$TBT_2 = \sum_i \left(\frac{L_{p,i}}{2V_p} \right) + \frac{(n-1)*h}{V_t} \quad (16)$$

For SMLP, traverse time will be equal to D/V_{max} , where D is the distance covered by the head from the end point of current layer to the next start point without any collisions. It is represented by V_{max} and will take less time to reach the start location. The total build time for m extruders can be calculated as,

$$TBT_3 = \left\lceil \frac{n}{2} \right\rceil_c * \frac{L_p}{V_p} + \left\lceil \frac{n-m}{2} \right\rceil_c * \frac{D}{V_{max}} + ((n+1)\%m * t_d) \quad (17)$$

where n is the number of layers, $\lceil . \rceil_c$ is the ceiling function and $\%$ is a modulo operation. In this work, as two extruders are used, the equation will become as,

$$TBT_3 = \left\lceil \frac{n}{2} \right\rceil_c * \frac{L_p}{V_p} + \left\lceil \frac{n-2}{2} \right\rceil_c * \frac{D}{V_{max}} + ((n+1)\%2 * t_d) \quad (18)$$

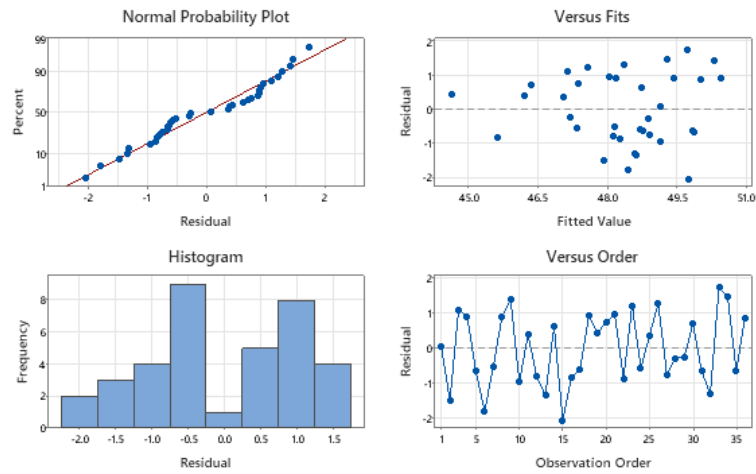


Figure 19 Residual plots for validating assumptions for ANOVA testing for tensile strength.

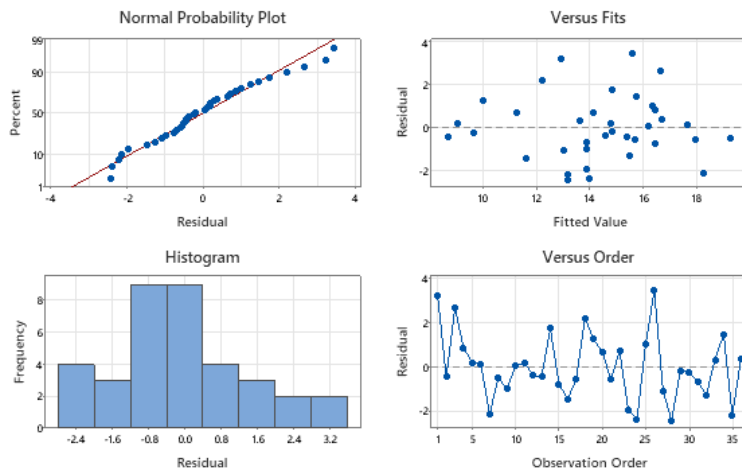


Figure 20. Residual plots for validating assumptions for ANOVA testing for tear strength.